

Sequences and Series of Numbers

4.1 Sequences of Numbers

4.1.1 THEOREM. (O. Stolz)¹-Cesaro²) If the sequences (x_n) and (y_n) satisfy at least one of the conditions:

- (1) (y_n) is strictly monotone, $x_n \rightarrow 0$, $y_n \rightarrow 0$;
- (2) (y_n) is strictly monotone, $|y_n| \rightarrow \infty$;

and if the limit

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n}$$

exists, then

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n}.$$

Proof.

4.1.2 THEOREM. (Ivan's Convergence Test for Sequences)

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of nonzero real numbers. If the limit

$$\lim_{n \rightarrow \infty} n^\alpha \left(\frac{x_{n+1}}{x_n} - 1 \right) = \lambda \in \mathbb{R} \setminus \{0\}$$

exists, then the limit $\lim_{n \rightarrow \infty} x_n$ also exists, and:

$$\lim_{n \rightarrow \infty} x_n = \begin{cases} 0, & \text{for } \alpha \leq 1, \quad \lambda < 0; \\ \pm\infty, & \text{for } \alpha \leq 1, \quad \lambda > 0; \\ \in \mathbb{R} \setminus \{0\}, & \text{for } \alpha > 1. \end{cases}$$

No proof. We strongly encourage you to learn by heart Theorem 4.1.2.

¹Otto Stolz (1842-1905), a German mathematician.

²Ernesto Cesaro (1859-1906), an Italian mathematician.

Test the following sequences for convergence:

4.1.3

$$x_n = \frac{a(a+1) \cdots (a+n-1) e^n \sqrt{n}}{n^{n+a}}, \quad n \geq 1, \quad a > 0.$$

4.1.4

$$y_n = \frac{n! n^{a-1}}{a(a+1) \cdots (a+n-1)}, \quad n \geq 1, \quad a > 0.$$

4.1.5

$$z_n = \frac{n! e^n}{n^n \sqrt{n}}, \quad n \geq 1.$$

4.1.6

$$t_n = \frac{\left(1 - \left(\frac{a}{2}\right)^2\right) \left(1 - \left(\frac{a}{4}\right)^2\right) \cdots \left(1 - \left(\frac{a}{2n}\right)^2\right)}{\left(1 - \left(\frac{a}{1}\right)^2\right) \left(1 - \left(\frac{a}{3}\right)^2\right) \cdots \left(1 - \left(\frac{a}{2n-1}\right)^2\right)}, \quad n \geq 1, \quad 0 < a < 1.$$

4.2 Series of Numbers

Let (x_n) be a sequence of complex numbers. Consider the sequence (S_n) ,

$$S_n = x_0 + \cdots + x_n, \quad (n \in \mathbb{N}).$$

The pair $((x_n), (S_n))$ is called a **series** and is denoted by

$$\sum x_n \quad \text{or} \quad \sum_{n \geq 0} x_n \quad \text{or} \quad \sum_{n=0}^{\infty} x_n.$$

The number S_n is called the ***n*-th partial sum** of the series and x_n is called the **general term** of the series.

If the sequence (S_n) has a finite limit as $n \rightarrow \infty$, then the series is said to be **convergent** and the limit is called the **sum** of series. In order to denote the sum of a series we shall use the same notations:

$$\sum x_n \quad \text{or} \quad \sum_{n \geq 0} x_n \quad \text{or} \quad \sum_{n=0}^{\infty} x_n,$$

that is

$$\sum_{n=0}^{\infty} x_n \stackrel{\text{def.}}{=} \lim_{n \rightarrow \infty} S_n.$$

If the sequence (S_n) has no limit, or the limit is infinite, then the series is said to be **divergent**.

The series

$$\sum_{k=n+1}^{\infty} x_k$$

is called the ***n*-th remainder series**.

A series of positive terms is referred to as a **positive series**. The sequence of the partial sums of a positive series is increasing and hence it has a limit; so we deduce

4.2.1 REMARK. A positive series is convergent iff the sequence of the partial sums of series is bounded.

4.2.2 DEFINITION. The series $\sum x_n$ is said to be **absolutely convergent** if the series $\sum |x_n|$ is convergent.

4.3 Convergence Tests for Series

4.3.1 THEOREM. (**Cauchy's Criterion**) A series $\sum x_n$ is convergent if and only if for any $\varepsilon > 0$ there exists r_ε such that the inequality

$$|x_{n+1} + \cdots + x_{n+p}| < \varepsilon$$

is satisfied for all $n, p \in \mathbb{N}$ and $n > r_\varepsilon$.

4.3.2 THEOREM. If a series $\sum x_n$ is convergent, then $\lim_{n \rightarrow \infty} x_n = 0$.

Let (u_n) and (v_n) be two sequences of positive terms.

4.3.3 THEOREM. (**First Comparison Test**) If $u_n \leq v_n$, for $n \in \mathbb{N}$, then:

- (1) $\sum u_n - \text{divergent} \implies \sum v_n - \text{divergent};$
- (2) $\sum v_n - \text{convergent} \implies \sum u_n - \text{convergent}.$

Proof.

4.3.4 THEOREM. (**Second Comparison Test**) If

$$\frac{u_{n+1}}{u_n} \leq \frac{v_{n+1}}{v_n},$$

for all $n \in \mathbb{N}$, then:

- (1) $\sum u_n - \text{divergent} \implies \sum v_n - \text{divergent};$
- (2) $\sum v_n - \text{convergent} \implies \sum u_n - \text{convergent}.$

Proof.

4.3.5 THEOREM. (**Third Comparison Test**) If

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = L \in (0, \infty),$$

then the series

$$\sum u_n \quad \text{and} \quad \sum v_n,$$

are simultaneously divergent or convergent.

Proof.

4.3.6 THEOREM. (**Cauchy's Root Test**)

- (1) If $\sqrt[n]{u_n} \leq q < 1$, for a certain index n_0 onwards, then $\sum u_n$ is convergent;
- (2) If $\sqrt[n]{u_n} \geq 1$, beyond a certain index n_0 , then $\sum u_n$ is divergent.

Proof.

4.3.7 COROLLARY. If $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell$, then:

$$\ell < 1 \implies \sum u_n - \text{convergent};$$

$$\begin{aligned}\ell > 1 &\Rightarrow \sum u_n - \text{divergent}; \\ \ell = 1 &\Rightarrow - \text{the test fails.}\end{aligned}$$

4.3.8 THEOREM. (D'Alembert Ratio Test)³

- (1) If $\frac{u_{n+1}}{u_n} \leq q < 1$, for a certain index n_0 onwards, then $\sum u_n$ is convergent;
- (2) If $\frac{u_{n+1}}{u_n} \geq 1$, beyond a certain index n_0 , then $\sum u_n$ is divergent.

Proof.

4.3.9 COROLLARY. If $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell$, then:

$$\begin{aligned}\ell < 1 &\Rightarrow \sum u_n - \text{convergent}; \\ \ell > 1 &\Rightarrow \sum u_n - \text{divergent}; \\ \ell = 1 &\Rightarrow - \text{the test fails.}\end{aligned}$$

4.3.10 THEOREM. (Kummer's Test)⁴ Let $\sum u_n$ be a series with positive terms and (c_n) be a sequence of positive real numbers. We have:

- (1) if there exist $r > 0$ and $p \in \mathbb{N}$ such that

$$c_n \frac{u_n}{u_{n+1}} - c_{n+1} \geq r,$$

$\forall n \geq p$, then $\sum u_n$ is convergent;

- (2) if there exists $p \in \mathbb{N}$ such that

$$c_n \frac{u_n}{u_{n+1}} - c_{n+1} \leq 0,$$

$\forall n \geq p$, and if

$$\sum \frac{1}{c_n}$$

is divergent, then $\sum u_n$ is divergent.

Proof.

If in Kummer's test we take $c_n = n$ we obtain

4.3.11 THEOREM. (Raabe⁵-Duhamel⁶'s Test)

- (1) if $\exists r > 1$ such that

$$n \left(\frac{u_n}{u_{n+1}} - 1 \right) \geq r,$$

$\forall n \in \mathbb{N}^*$, then the series $\sum u_n$ is convergent;

- (2) if for all $n \in \mathbb{N}^*$ we have

$$n \left(\frac{u_n}{u_{n+1}} - 1 \right) \leq 1,$$

then the series $\sum u_n$ is divergent.

4.3.12 COROLLARY. If $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \ell$, then:

³Jean le Rond d'Alembert (1717-1783), a French mathematician.

⁴Ernst Eduard Kummer (1810-1893), a German mathematician.

⁵Joseph Ludwig Raabe (1801-1859), a Swiss mathematician.

⁶Jean Marie Constant Duhamel (1797-1872), a French mathematician.

$$\begin{aligned}\ell > 1 &\Rightarrow \sum u_n - \text{convergent}; \\ \ell < 1 &\Rightarrow \sum u_n - \text{divergent}; \\ \ell = 1 &\Rightarrow - \text{the test fails.}\end{aligned}$$

4.3.13 THEOREM. (Associativity Property) Let $\sum a_n$ be a series with positive terms and (k_n) be a strictly increasing sequence of positive integers, $k_0 = 0$. If

$$b_0 = a_0, \quad b_n = a_{k_{n-1}+1} + a_{k_{n-1}+2} + \cdots + a_{k_n},$$

$n = 1, 2, \dots$, then the series $\sum a_n$ and $\sum b_n$ are either divergent or convergent.

Proof.

4.3.14 THEOREM. (Cauchy's Condensation Test) If (a_n) is a decreasing sequence of positive terms, and (k_n) is an increasing sequence of positive integers, such that

$$\frac{k_{n+2} - k_{n+1}}{k_{n+1} - k_n} = O(1),$$

$n \in \mathbb{N}$, then both series

$$\sum a_n \quad \text{and} \quad \sum (k_{n+1} - k_n) a_{k_n}$$

are either divergent or convergent.

Proof. Denoting

$$b_n = a_{k_n+1} + \cdots + a_{k_{n+1}},$$

we obtain:

$$(k_{n+1} - k_n) a_{k_{n+1}} \leq b_n \leq (k_{n+1} - k_n) a_{k_n},$$

$n \geq 0$. There exists $M > 0$ such that

$$0 \leq \frac{k_{n+2} - k_{n+1}}{k_{n+1} - k_n} \leq M,$$

therefore

$$\frac{k_{n+2} - k_{n+1}}{M} a_{k_{n+1}} \leq b_n \leq (k_{n+1} - k_n) a_{k_n},$$

$n \geq 0$, and using the first comparison test (4.3.3), we deduce that both series

$$\sum b_n, \quad \sum (k_{n+1} - k_n) a_{k_n}$$

are either divergent or convergent.

By theorem (4.3.14), the series $\sum b_n$ and $\sum a_n$ are simultaneously divergent or convergent, therefore the series

$$\sum a_n \quad \text{and} \quad \sum (k_{n+1} - k_n) a_{k_n}$$

are simultaneously divergent or convergent. ◇◇◇

For $k_n = 2^n$ we obtain:

4.3.15 COROLLARY. If (a_n) is a decreasing sequence of positive terms, then the series

$$\sum a_n \quad \text{and} \quad \sum 2^n a_{2^n}$$

are simultaneously divergent or convergent.

4.3.16 EXAMPLE. Examine the convergence of the generalized harmonic series

$$\sum_{n \geq 1} \frac{1}{n^\alpha}.$$

For $\alpha \leq 0$ the sequence $\frac{1}{n^\alpha}$ does not converge to zero, therefore the series is divergent.

For $\alpha > 0$, using (4.3.15) we deduce that the series has the same behaviour as the infinite geometric progression

$$\sum_{n \geq 1} 2^n \left(\frac{1}{2^n} \right)^\alpha = \sum_{n \geq 1} \left(\frac{1}{2^{\alpha-1}} \right)^n,$$

which is convergent for $\frac{1}{2^{\alpha-1}} < 1$. Consequently,

$$\sum_{n \geq 1} \frac{1}{n^\alpha} \text{ is } \begin{cases} \text{convergent,} & \text{for } \alpha > 1, \\ \text{divergent,} & \text{for } \alpha \leq 1. \end{cases}$$

4.3.17 EXAMPLE. Investigate the following series for convergence

$$\sum_{n \geq 2} \frac{1}{n \ln n}.$$

Using (4.3.15) and the fact that the series

$$\sum_{n \geq 2} 2^n \frac{1}{2^n \ln 2^n} = \frac{1}{\ln 2} \sum_{n \geq 2} \frac{1}{n}$$

is divergent we deduce that the series is divergent.

4.3.18 THEOREM. (**Abel⁷-Dirichlet⁸ Test**) Let the terms of the sequence (a_n) decrease monotonically to zero and the partial sum (S_n) of the series $\sum u_n$ be bounded. Then the series $\sum u_n a_n$ is convergent.

Proof. Denote:

$$\sigma_n = a_0 u_0 + a_1 u_1 + \cdots + a_n u_n.$$

We have:

$$\begin{aligned} \sigma_{n+p} - \sigma_n &= a_{n+1} u_{n+1} + \cdots + a_{n+p} u_{n+p} = \\ &= a_{n+1}(S_{n+1} - S_n) + a_{n+2}(S_{n+2} - S_{n+1}) + \cdots + a_{n+p}(S_{n+p} - S_{n+p-1}) = \\ &= -a_{n+1} S_n + S_{n+1}(a_{n+1} - a_{n+2}) + \cdots + S_{n+p-1}(a_{n+p-1} - a_{n+p}) + a_{n+p} S_{n+p}, \end{aligned}$$

therefore, we deduce

$$\begin{aligned} |\sigma_{n+p} - \sigma_n| &\leq M(a_{n+1} + a_{n+p}) + M(a_{n+1} - a_{n+2} + \cdots + a_{n+p-1} - a_{n+p}) = \\ &= 2Ma_{n+1}. \end{aligned}$$

The inequalities

$$|\sigma_{n+p} - \sigma_n| \leq 2Ma_{n+1}, \quad n, p \in \mathbb{N},$$

and the fact that the sequence (a_n) converges to zero show that the sequence (σ_n) is a Cauchy sequence, and consequently, it is convergent; so we deduce the convergence of the series $\sum u_n a_n$. $\diamond\diamond\diamond$

4.3.19 THEOREM. (**Leibniz's⁹ Test**) If the sequence (a_n) decreases to zero, then the alternating series $\sum (-1)^n a_n$ is convergent.

Proof. Denote:

$$S_n = 1 - 1 + 1 - \cdots + (-1)^n.$$

⁷Niels Henrik Abel (1802-1829), a Norwegian mathematician.

⁸Peter Gustave Lejeune Dirichlet (1805-1859), a German mathematician.

⁹Gottfried Wilhelm Leibniz (1646-1716), a great German mathematician.

Observe that the sequence (S_n) is bounded. By virtue of theorem (4.3.18) the series $\sum (-1)^n a_n$ is convergent. $\diamond \diamond$

4.3.20 EXAMPLE. Investigate the series $\sum \frac{\sin n}{\ln(n+1)}$ for convergence.

We have:

$$S_n = \sin 1 + \cdots + \sin n = \frac{\sin \frac{n+1}{2} \sin \frac{n}{2}}{\sin \frac{1}{2}},$$

hence we deduce

$$|S_n| \leq \frac{1}{\sin \frac{1}{2}}, \quad n \in \mathbb{N}^*,$$

and consequently, using the result $\frac{1}{\ln(n+1)} \rightarrow 0$, we deduce that the series $\sum \frac{\sin n}{\ln(n+1)}$ is convergent.

4.4 Infinite Products

Consider a sequence (a_n) of real numbers different from zero. Define the sequence (P_n) ,

$$P_n := a_0 \cdots a_n, \quad n \in \mathbb{N}.$$

The pair $((a_n), (P_n))$ is called an **infinite product** and is denoted by

$$\prod a_n \quad \text{or} \quad \prod_{n \geq 0} a_n \quad \text{or} \quad \prod_{n=0}^{\infty} a_n.$$

If the sequence (P_n) is convergent, then its limit is denoted by the same notations:

$$\prod_{n \geq 0} a_n \quad \text{or} \quad \prod_{n=0}^{\infty} a_n.$$

If $\lim_{n \rightarrow \infty} P_n = 0$ we say that $\prod a_n$ diverges to zero.

If (P_n) is convergent to a number different from zero, then $\prod a_n$ is convergent.

Let (a_n) be a sequence of positive numbers. Using the identity

$$\ln(a_0 \cdots a_n) = \ln a_0 + \cdots + \ln a_n,$$

we deduce that

$$\prod a_n \text{ — convergent} \iff \sum \ln a_n \text{ — convergent},$$

furthermore

$$\ln \left(\prod_{n \geq 0} a_n \right) = \sum_{n \geq 0} \ln a_n.$$

4.4.1 THEOREM. If (a_n) is a sequence of positive numbers, then both the product $\prod(1 + a_n)$ and the series $\sum a_n$ are either divergent or convergent.

Proof. The series $\sum a_n$ and $\sum \ln(1 + a_n)$ are positive series. If $\sum a_n$ is convergent then $a_n \rightarrow 0$, hence

$$\frac{\ln(1 + a_n)}{a_n} \rightarrow 1,$$

therefore $\sum \ln(1 + a_n)$ is convergent by the third comparison test (4.3.5). Using the fact that the series $\sum \ln(1 + a_n)$ and the product $\prod(1 + a_n)$ are simultaneously divergent or convergent, we deduce that the product $\prod(1 + a_n)$ is convergent.

Similarly, we can show that if the product $\prod(1 + a_n)$ is convergent, then the series $\sum a_n$ is also convergent.